

Adrenal Glands and Cortisol

Cortisol and the stress response (NR3C1, FKBP5), cortisol transport (SERPINA6), steroid synthesis in the gland (CYP11B1/B2, DHEA-S) and adrenal autoimmunity (Addison's disease)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Adrenal autoimmunity (Addison's disease) has a low genetic predisposition: the burden of 10 replicated loci is at percentile **6.5**. Sérgio F does not carry the risk allele of **HLA-DQ** (C/C), the largest genetic factor for the disease (about 6×), nor those of **AIRE** (C/C), **PTPN22 R620W** (G/G), LPP and CTLA4. In the stress axis, **FKBP5** lacks the slow cortisol-recovery alleles (rs1360780 C/C, rs3800373 A/A, rs9470080 C/C) and **NR3C1 9β** is absent (T/T), which favours a glucocorticoid receptor without relative cortisol resistance.

Unfavourable. There is 1 risk allele in **BACH2** (C/A), the second locus for Addison's disease (about 1.7×), which counts for little on its own. Sérgio F does not carry the protective **ER22/23EK** allele of NR3C1 (C/C) or the resilience allele of CRHR1 (G/G); these are the genotypes most people have.

No judgement (descriptive). The adrenal androgens have genetics for higher values: the burden of 43 **DHEA-S** loci is at percentile **90.0**, with 2 raising alleles in SULT2A1 and CYP3A7. Blood cortisol has typical genetics (burden at percentile 32.7; SERPINA6 without the alleles that raise cortisol-binding globulin). In steroid synthesis, CYP11B2 has 2 copies of the –344T allele (more aldosterone synthase activity in the gland) and CYP11B1 has 1 allele; the aldosterone loci burden is at percentile 44.2.

What it means. The adrenal glands have a calm genetic profile: gland autoimmunity is unlikely, and the stress axis (FKBP5 and NR3C1) is genetically good at switching off the cortisol response. The genetically higher DHEA-S is a value to compare with blood tests; DHEA-S falls considerably with the years and higher values are associated with better health. The cortisol axis depends mainly on sleep, light timing, chronic stress and training.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality $DR2 \geq 0.8$. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 20 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 15 from the chip, 5 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 3 SNPs needed conversion to the complementary strand and 1 is palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 1 PGS Catalog score calculated locally with 1 variant in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 75 loci, 65 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Addison's disease	—	0 of 1	—	0

There is only one published score for Addison's disease (14 variants) and its HLA variants were not read; therefore there is no consensus, and the reading of this gland rests on the loci burden and the key variants.

Addison's disease

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Autoimmune Addison's disease	PGS004239 (Aranda-Guillén 2023, 14 variants)	1 of 11 (coverage 6%)	40.1	TYPICAL	typical (low reliability)

- **Autoimmune Addison's disease:** Only 1 of the 14 variants was read (coverage 6%): the score's HLA class II variants are not on the chip or in the good-quality imputation. The result cannot be interpreted; the Addison's disease loci burden replaces it.

4. Key variants by topic

Variants of the glucocorticoid receptor (NR3C1) and its regulator FKBP5, of cortisol transport and inactivation, of the gland's steroid synthesis enzymes (CYP11B1, CYP11B2, SULT2A1) and of adrenal autoimmunity. Genes with only a quantitative effect (hormone levels) receive no judgement.

Glucocorticoid receptor and stress response

NR3C1 is the cortisol receptor; FKBP5 restrains it and regulates how the stress response is switched off.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NR3C1	rs6190	ER22/23EK	C/C	chip	A (0 copies)	neutral	ER22/23EK (R23K): the A allele makes the receptor less sensitive to cortisol; it is associated with better metabolic profile and less insulin resistance	Moderate
NR3C1	rs6198	9β A3669G	T/T	imputed DR2=1.00	G (0 copies)	favourable	9β (A3669G): the G allele stabilises GRβ and gives a relative resistance to cortisol; more basal inflammation	Weak
FKBP5	rs1360780		C/C	chip	T (0 copies)	favourable	FKBP5 intron 2: the T allele increases the induction of FKBP5 by cortisol and delays the closure of the stress axis (especially with early stress)	Moderate
FKBP5	rs3800373		A/A	chip	C (0 copies)	favourable	FKBP5 3'UTR: the C allele follows rs1360780 T (slow cortisol recovery haplotype)	Weak
FKBP5	rs9470080		C/C	chip	T (0 copies)	favourable	FKBP5: the T allele is associated with slower cortisol recovery after stress	Weak
CRHR1	rs110402		G/G	chip	A (0 copies)	neutral	CRH receptor 1: the A allele is associated with more resilience to early stress	Weak

Cortisol transport and metabolism

Cortisol-binding globulin (SERPINA6) and the enzymes that regenerate or inactivate cortisol.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SERPINA6	rs9989237		C/C	imputed DR2=0.99	T (0 copies)	no judgement	Cortisol-binding globulin (CBG): the largest locus of morning cortisol in blood [GWAS Catalog: Plasma cortisol levels (morning), allele T, beta 0.091 [0.074-0.108] unit increase, p=1E-26, PMID 38525495]	Strong
SERPINA6	rs12589136		G/G	chip	T (0 copies)	no judgement	SERPINA6/SERPINA1 region (CBG): second signal for morning cortisol [GWAS Catalog: Cortisol levels, allele T, beta 0.138 [0.1-0.18] unit increase, p=1E-13, PMID 35050183]	Moderate
HSD11B1	rs12086634		T/G	chip	G (1 copy)	no judgement	11β-HSD1 (regenerates cortisol from cortisone in liver and fat): the G allele is associated with less activity	Weak

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
AKR1D1	rs3857868		A/A	imputed DR2=0.99	A (2 copies)	no judgement	Hepatic 5 β -reductase (inactivation of cortisol) [GWAS Catalog: Tetrahydrocortisol glucuronide levels, allele A, beta 0.138 [0.1-0.17] unit increase, p=3E-16, PMID 36635386]	Weak

Steroid synthesis in the gland

Final steps of cortisol and aldosterone synthesis (CYP11B1/B2) and DHEA sulphation (SULT2A1).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP11B2	rs1799998	–344C/T	A/A	chip	T (2 copies)	no judgement	Aldosterone synthase (–344C/T): the T allele is associated with more aldosterone production in the gland	Weak
CYP11B1	rs6392		C/T	imputed DR2=1.00	T (1 copy)	no judgement	CYP11B1 region (11 β -hydroxylase, last step of cortisol synthesis in the gland): associated with sex steroid levels [GWAS Catalog: Total testosterone levels, allele C, beta 0.020 [0.014-0.026] nmol/L decrease; risk/increase allele = T, the opposite, p=1E-11, PMID 34321204]	Weak
SULT2A1	rs2637125		G/G	chip	G (2 copies)	no judgement	DHEA sulphotransferase (DHEA-S, the main adrenal androgen) [GWAS Catalog: Dehydroepiandrosterone sulphate levels, allele A, beta 0.090 [0.07-0.11] umol/L decrease; risk/increase allele = G, the opposite, p=3E-19, PMID 21533175]	Moderate
ZKSCAN5/ CYP3A7	rs11761528		C/C	chip	C (2 copies)	no judgement	CYP3A7*1C region: metabolism of DHEA-S [GWAS Catalog: Dehydroepiandrosterone sulphate levels, allele T, beta 0.160 [0.14-0.18] umol/L decrease; risk/increase allele = C, the opposite, p=3E-36, PMID 21533175]	Moderate

Adrenal autoimmunity (Addison's disease)

Immune tolerance genes shared with other endocrine autoimmune diseases.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BACH2	rs10806425		C/A	chip	A (1 copy)	unfavourable (mild)	BACH2 (regulation of T lymphocytes): ~1.7 $\times$ more autoimmune Addison's disease per copy	Strong
AIRE	rs74203920		C/C	chip	T (0 copies)	favourable	AIRE (central thymus tolerance): large-effect variant in Addison's disease (~3.4 $\times$)	Strong
PTPN22	rs2476601	R620W	G/G	chip	A (0 copies)	favourable	R620W: ~1.7 $\times$ more Addison's disease and other autoimmunity	Strong
LPP	rs1464510		C/C	chip	A (0 copies)	favourable	LPP: ~1.4 $\times$ more Addison's disease	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CTLA4	rs231775	+49A/G	A/A	chip	G (0 copies)	favourable	CTLA4 +49A/G: the G allele is associated with more endocrine autoimmunity	Weak
HLA-DQA1/DQB1	rs3998178		C/C	imputed DR2=1.00	T (0 copies)	favourable	HLA-DQ region (DR3-DQ2 and DR4-DQ8 haplotypes): the largest genetic factor of Addison's disease (~6×)	Strong

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for cortisol, DHEA-S, aldosterone, Addison's disease and adrenal tumours. For hormone levels, alleles are oriented towards higher levels (no judgement); for diseases, towards more risk.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Cortisol (levels)	6 (6)	2 of 12	6	32.7	no judgement
DHEA-S (adrenal androgen)	47 (43)	64 of 86	43	90.0	no judgement
Aldosterone and aldosterone-producing adenoma	4 (4)	2 of 8	4	44.2	no judgement
Addison's disease	10 (10)	4 of 20	10	6.5	favourable
Adrenal tumours and hyperfunction	8 (2)	0 of 4	2	9.4	favourable

6. Favourable

- **Autoimmune Addison's disease:** loci burden at percentile **6.5**; HLA-DQ (C/C), AIRE, PTPN22 R620W, LPP and CTLA4 without risk alleles. **Moderate to strong.**
- **FKBP5 without the slow-recovery haplotype** (rs1360780 C/C, rs3800373 A/A, rs9470080 C/C): cortisol comes down faster after stress. **Moderate.**
- **NR3C1 9β absent** (T/T): without the relative cortisol resistance associated with more baseline inflammation. **Weak.**
- **Adrenal tumours and hyperfunction:** burden at percentile 9.4, but with only 2 loci read. **Weak.**

7. Unfavourable

- **BACH2** C/A: 1 risk allele for Addison's disease (about 1.7×); no meaning on its own given the low total burden. **Weak.**
- **NR3C1 ER22/23EK absent** (C/C) and **CRHR1** without the resilience allele (G/G): common genotypes, without the metabolic and resilience advantage these alleles give. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Endocrine Dysregulation	Cortisol axis: FKBP5 and NR3C1 favourable; typical genetic cortisol; genetically higher DHEA-S (loci at percentile 90.0)	Favourable
Chronic Inflammation	Cortisol is the main endogenous anti-inflammatory hormone; without the 9β cortisol-resistance variant	Favourable
Altered Intercellular Communication	Hormonal signalling between the pituitary and the adrenal (HPA axis) without slow-recovery variants	Favourable
Circadian Disruption	Cortisol follows the circadian clock (morning peak); the genetics is typical and the clock depends mainly on light and sleep	Conditional
Insulin Resistance	Excess cortisol action favours insulin resistance; no hypersensitivity variants read	Typical
Immunosenescence	Endocrine autoimmunity unlikely (Addison loci at percentile 6.5)	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **DHEA-S in routine blood tests** (once), to compare with the genetic predisposition to higher values. Do not supplement DHEA without a confirmed low value and a medical discussion. *Suggestion (not applied)*.
2. **Protect the cortisol rhythm**: daylight in the morning, regular sleep times and little light at night. Melatonin 0.3–1 mg is already in the plan and does not change the genetics of the axis. *Suggestion (not applied)*.
3. **Corticosteroids**: if corticosteroids are ever needed for more than 3 weeks, remind the doctor of the risk of adrenal suppression and of the need for a gradual taper. Genetics does not change this rule. *Suggestion (not applied)*.
4. **Signs of Addison's disease** (intense fatigue, darkening of the skin, salt craving, dizziness on standing): unlikely on genetic grounds, but they justify a morning cortisol and ACTH if they appear. *Suggestion (not applied)*.
5. **Already in the plan and protective**: daily exercise, sleep and glucose control. *Suggestion (not applied)*.

10. Limitations

- There is only one polygenic score for Addison's disease and its HLA variants were not read; the reading rests on the loci burden and the key variants.
- Cortisol, aldosterone and DHEA-S levels have few known loci and vary a lot with time of day, stress and sleep; genetics explains a small part.
- Rare adrenal diseases (congenital adrenal hyperplasia due to CYP21A2, and MC2R, STAR and NR0B1 mutations) are not read by the chip.
- The effect of FKBP5 depends strongly on early-life stress; without that context the effect is small.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; where the literature uses the gene strand (NR3C1, CYP11B2), the letters appear complemented. Published alleles with a lowering effect were inverted to the raising direction.
- **Polygenic scores:** GRCh38 harmonised file from the PGS Catalog (Addison's disease). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 5 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Eriksson 2021 and Aranda-Guillén 2023 (Addison's disease GWAS and score), van Rossum 2002 and Manenschijn 2009 (NR3C1), Binder 2008 (FKBP5), Crawford 2021 (cortisol and SERPINA6), Zhai 2011 (DHEA-S), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Glucocorticoid receptor and stress response	NR3C1	rs6190	C/C	chip	A	0	complement (– strand)	without the protective allele (common genotype)
Glucocorticoid receptor and stress response	NR3C1	rs6198	T/T	imputed DR2=1.00	G	0	complement (– strand)	without the risk allele
Glucocorticoid receptor and stress response	FKBP5	rs1360780	C/C	chip	T	0	+ strand	without the risk allele
Glucocorticoid receptor and stress response	FKBP5	rs3800373	A/A	chip	C	0	+ strand	without the risk allele
Glucocorticoid receptor and stress response	FKBP5	rs9470080	C/C	chip	T	0	+ strand	without the risk allele
Glucocorticoid receptor and stress response	CRHR1	rs110402	G/G	chip	A	0	+ strand	without the protective allele (common genotype)
Cortisol transport and metabolism	SERPINA6	rs9989237	C/C	imputed DR2=0.99	T	0	+ strand	0 copies of the effect allele
Cortisol transport and metabolism	SERPINA6	rs12589136	G/G	chip	T	0	+ strand	0 copies of the effect allele
Cortisol transport and metabolism	HSD11B1	rs12086634	T/G	chip	G	1	+ strand	1 copy of the effect allele
Cortisol transport and metabolism	AKR1D1	rs3857868	A/A	imputed DR2=0.99	A	2	+ strand (palindromic; + assumed)	2 copies of the effect allele
Steroid synthesis in the gland	CYP11B2	rs1799998	A/A	chip	T	2	complement (– strand)	2 copies of the effect allele
Steroid synthesis in the gland	CYP11B1	rs6392	C/T	imputed DR2=1.00	T	1	+ strand	1 copy of the effect allele
Steroid synthesis in the gland	SULT2A1	rs2637125	G/G	chip	G	2	+ strand	2 copies of the effect allele
Steroid synthesis in the gland	ZKSCAN5/ CYP3A7	rs11761528	C/C	chip	C	2	+ strand	2 copies of the effect allele
Adrenal autoimmunity (Addison's disease)	BACH2	rs10806425	C/A	chip	A	1	+ strand	1 copy of the risk allele
Adrenal autoimmunity (Addison's disease)	AIRE	rs74203920	C/C	chip	T	0	+ strand	without the risk allele
Adrenal autoimmunity (Addison's disease)	PTPN22	rs2476601	G/G	chip	A	0	+ strand	without the risk allele
Adrenal autoimmunity (Addison's disease)	LPP	rs1464510	C/C	chip	A	0	+ strand	without the risk allele
Adrenal autoimmunity (Addison's disease)	CTLA4	rs231775	A/A	chip	G	0	+ strand	without the risk allele
Adrenal autoimmunity (Addison's disease)	HLA-DQA1/ DQB1	rs3998178	C/C	imputed DR2=1.00	T	0	+ strand	without the risk allele